

VIVEK JAISWAL

Senior Software Engineer — Python | Django | FastAPI | Generative AI & Agentic AI

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JOB OBJECTIVE

Driving the development of scalable, AI-powered, and high-performance applications by leveraging 4.5+ years of expertise in **Python, Django, FastAPI, Flask, Machine Learning, NLP, Generative AI**, and **Agentic AI** technologies. Seeking a Senior/Lead Software Engineer role to contribute to intelligent backend systems, autonomous LLM agents, **LangGraph** workflows, **Model Context Protocol (MCP)** servers, and enterprise-scale, cloud-native data platforms.

CORE COMPETENCIES

- **Python, Django, FastAPI & Flask** Development
- RESTful & Async API Design
- **Generative AI, LLMs, LangChain & LLMops**
- **Microservices Architecture & System Design** (HLD/LLD)
- Database Design & Query Optimization (ORMs)
- **Asynchronous & Concurrent** Systems
- Background Jobs & Workflow Orchestration (**Celery**)
- DevOps, **CI/CD** & Infrastructure Automation
- **Machine Learning** Integration in APIs
- **Full-Text Search & Index** Tuning (**Elasticsearch**)
- **Agentic AI & Multi-Agent Workflows** (**LangGraph**)
- **Model Context Protocol (MCP)** Integration & Custom Servers
- **Retrieval-Augmented Generation (RAG) & Vector Databases**
- Event-Driven Systems & Messaging (**Kafka**)
- **Distributed Systems & High-Availability** Architecture
- Caching & Performance Tuning (**Redis**)
- **Cloud-Native** & Containerized Deployments (**Docker/K8s**)
- **Unit Testing & TDD** (**Pytest**)
- Graph Databases & **Knowledge Graph** Modeling (**Neo4j**)
- **ETL Pipeline** Design & **Data Engineering**

PROFILE SUMMARY

Results-driven Senior Software Engineer with 4.5+ years of experience designing scalable backend systems and AI-powered applications using **Python, Django, DRF, FastAPI**, and **Flask**. Strong expertise in **Microservices Architecture, Distributed Systems, Asynchronous** processing, **System Design**, and cloud-native backend development, with hands-on experience integrating **Machine Learning, Generative AI**, and **Agentic AI** solutions into enterprise applications.

Experienced in designing and deploying **NLP** and **LLM**-powered systems using **Hugging Face Transformers, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Prompt Engineering, Semantic Search**, and **Vector Embeddings**, with working exposure to **LLMops** for deploying and monitoring **LLM**-driven services. Strong experience in **Data Engineering** using **PySpark, Kafka, Redis**, and **Elasticsearch**, with hands-on expertise in **AWS, Docker, Kubernetes, Neo4j, TensorFlow**, and **MLOps** workflows for production-grade AI systems.

TECHNICAL SKILLS

Languages: **Python**

Frameworks: **Django, DRF, Flask, FastAPI, SQLAlchemy, Celery, Hugging Face Transformers, LangChain, LangGraph**

Databases: **MySQL, PostgreSQL, MongoDB, Neo4j**, ORM Libraries, Query Optimization

Messaging & Distributed Systems: **Redis, Celery, Kafka, Zookeeper, Apache Spark (PySpark)**

Search & Vector Platforms: **Elasticsearch, Semantic Search, Text Embeddings, Vector Databases** (FAISS, Chroma, PGVector)

Generative AI & Agentic AI: **Agentic AI, LangGraph, Model Context Protocol (MCP), Generative AI, LLMs, NLP, Prompt Engineering, RAG, AI Agents, Conversational AI, Document Processing, LLMops**

ML & Data Science: **Machine Learning, Deep Learning, TensorFlow, Pandas, PySpark, NumPy, OpenCV, DeepFace, Computer Vision, ETL Pipelines**

DevOps & Cloud: **Docker, Kubernetes, AWS** (EC2, S3, Lambda, RDS, SNS, SQS, CloudWatch, API Gateway, IAM, EventBridge, ECR, ECS), **CI/CD**

Testing, Tools & APIs: **Unit Testing, Pytest, Postman, Git, Swagger, FCM, Google Dialogflow**

Architecture: **REST API Design, System Design, Scalable Architecture, Cron Jobs, Push Notifications**

ACHIEVEMENTS

- **HackerRank Certifications:** Earned 5-star rating in **Python** and 5-star rating in **SQL**, showcasing strong proficiency in programming and database management.
- **Scalable System Design:** Designed and implemented a modular **Microservices**-based back-end architecture capable of supporting high-traffic enterprise applications with independent service deployment, horizontal scalability, and fault isolation.
- **Generative AI Development:** Built and integrated AI-powered applications leveraging **Hugging Face Transformers, NLP** pipelines, **Prompt Engineering, LangChain**, and **LangGraph** multi-agent **LLM** workflows for intelligent document understanding and conversational capabilities.
- **AI Innovation:** Designed scalable AI-enabled backend systems supporting **Semantic Search**, intelligent automation, custom **Model Context Protocol (MCP)** servers, and **Retrieval-Augmented Generation (RAG)** architecture with **Vector Databases**.
- **Dual Expertise:** Combined strong Backend Engineering with **Machine Learning** proficiency, integrating ML models into production-ready **Django/FastAPI** applications to enhance predictive functionality and user experience.
- **Professional Growth:** Progressed to managing high-impact and complex projects, reflecting trust in technical expertise and reliability.

WORK EXPERIENCE

Jan'26 – Present | Senior Software Engineer | Innefu Labs, New Delhi

- Designing and developing high-performance backend systems using **Python, Django, FastAPI**, and **Flask** microservices for internal tooling and AI-agent utility endpoints, following clean, **Microservices** architecture principles.

- Building RESTful and **Asynchronous** APIs using **FastAPI** (async/await) and **Django** for secure, scalable, low-latency services; maintaining **ETL** pipelines for structured and semi-structured data ingestion.
- Implementing graph-based data modeling using **Neo4j** and tuning **Elasticsearch** indexing/**Full-Text Search** for high-speed analytics and search-driven features, backed by **System Design** principles.
- Designing and developing autonomous **AI Agents** using **LangGraph** to coordinate multi-agent teams, managing state transitions, persistent chat memory, and self-correction loops.
- Architecting custom **Model Context Protocol (MCP)** servers to securely connect **LLM** agents to local data stores, graph databases, and external tool endpoints.
- Implementing **Retrieval-Augmented Generation (RAG)** using **Vector Embeddings** and **Semantic Search** for AI-powered knowledge retrieval, and integrating **LangChain** orchestration for prompt-based automation workflows.
- Contributing to core product architecture involving scalable backend systems, Graph Databases (**Neo4j**), and search-driven architectures (**Elasticsearch**) for high-performance, intelligent querying.

May'22 – Dec'25 | Back-end Developer | Shyam Future Tech Pvt. Ltd., Kolkata (Group of Shyam Steel)

- Designed, developed, and maintained scalable **RESTful APIs** using **Django**, **DRF**, **FastAPI**, and **Flask** — including standalone **Flask** microservices for internal automation utilities and third-party integrations — following **Microservices** and **Event-Driven Architecture** principles.
- Built **FastAPI** and **Flask** services with **SQLAlchemy** for ORM-based database management, schema migrations, and **Async** APIs using **async/await**, dependency injection, and connection pooling.
- Optimized backend performance using **Django** ORM, **SQLAlchemy**, middleware, serializers, query optimization, indexing, and profiling techniques.
- Integrated **Redis** for caching, rate limiting, and session management, and **Celery** for distributed task queues and scheduled/background jobs.
- Built and maintained Apache **Kafka** (cloud-managed & self-hosted) with Zookeeper for real-time data streaming and distributed event processing.
- Implemented **AWS** services (Lambda, SNS, SQS, EC2, S3, RDS, ECS, ECR, CloudWatch) for serverless processing, messaging, monitoring, and scalable infrastructure.
- Implemented **Elasticsearch** for high-performance indexing and **Full-Text Search**; managed large datasets using **MySQL**, **PostgreSQL**, and **MongoDB** with schema/query optimization.
- Containerized applications using **Docker** and deployed **Microservices** on **Kubernetes**, managing deployments, configs, secrets, and rolling updates.
- Used **PySpark** and Spark SQL for distributed processing of large datasets (100GB+) across multi-node clusters integrated with **Kafka**, S3, and relational databases.
- Integrated **Machine Learning** and Computer Vision pipelines using **TensorFlow**, **OpenCV**, DeepFace, **Pandas**, and **NumPy** for intelligent automation and predictive processing.
- Boosted API performance and scalability via ORM query optimization, **Redis** caching, and **Asynchronous FastAPI** services, achieving 99.9% uptime on a **Cloud-Native AWS** architecture (EC2, Lambda, SQS, SNS, **Docker**, **Kubernetes**).

Sept'17 – Dec'17 | Associate Technical Project Manager | NS Matrix Services Pvt. Ltd., New Delhi

- Designed and developed scalable **Django & Flask REST APIs**, architecting modular back-end services using OOP principles, design patterns, and clean architecture.
- Implemented advanced **Django** ORM and database optimizations (indexing, joins, annotations, prefetch/select_related, stored procedures) to support high-volume transactional workloads.
- Integrated **Redis** caching and **Celery**-based **Asynchronous** tasks to improve API performance and automate background workflows; deployed applications on **AWS** (EC2, RDS, S3) with **CI/CD** pipelines and **Docker** containers.

June'17 – Sept'17 | Technical Trainee | NS Matrix Services Pvt. Ltd., New Delhi

- Assisted in developing **Django** and **Flask**-based **REST APIs**, learning modular backend design using OOP principles and MVC/MVT architecture; gained hands-on experience with **Django** ORM and query optimization.

EDUCATION

Bachelor of Technology (B.Tech.) in Computer Science and Engineering

GLA University, Mathura (U.P.) | 2017 | CGPA: 7.61

SOFT SKILLS

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| • Analytical Problem-Solving | • Systematic Thinking |
| • Technical Communication | • Proactive Initiative |
| • Mentorship & Knowledge Sharing | • Stakeholder Collaboration |